B.Tech. BCSE313L - Fundamentals of Fog and Edge Computing

RULE COVERAGE DATA DIODE

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Programme

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Abstract

Sending less sensor data saves transmission capacity, but it can also remove the readings needed to answer monitoring rules. This report explores that problem using CS²-Collector, the compressive sensing method proposed by Wang et al. (2016). Their study measures how well compressed readings reconstruct an environmental signal. It does not establish whether the reduced data preserves threshold and count decisions when the receiver cannot ask for missing evidence. Rule Coverage Data Diode (RCDD) takes up this question. Before reducing raw data, the gateway records the facts required by registered rules and sets aside capacity to send them. Signed Rule Coverage Capsules let Fog return TRUE, FALSE or UNKNOWN based on the evidence received. This process runs through the project's twelve research phases, N0-N11. [1, P1-P3]

An offline pilot uses 8,709 unchanged records from five motes in the public Intel Lab dataset, which also supplied the reference study. In a 30× modeled replay, exact rule facts produce no wrong decisions across 6,336 rule-window opportunities. Withholding window evidence gives 1,276 UNKNOWN results, while a deliberately unsafe control based on window means gives 63 wrong decisions. Timing also matters: at the natural sampling cadence, 12,800 of 15,840 opportunities remain UNKNOWN. These results support keeping exact rule facts for real observations. The pilot does not reproduce CS²-Collector or test RCDD signing, transport or byte reservations, so it cannot show a transmission-cost advantage. [E1]

Keywords Edge- Fog computing; telemetry reduction; registered rule evidence; one way transfer; Intel Lab sensor data

Reference study

Wang et al. 2016 CS²-Collector Sensors 16(8), 1318

DOI 10.3390/s16081318

1 Introduction and Relevant Works

Sensor systems often cannot send everything they collect. Compression and aggregation help them stay within a link's capacity, but they may remove a reading that a monitoring application needs. This project asks what Fog can still conclude once the Edge gateway has accepted the observations and discarded some of the raw detail.

1.1 Reference study and research question

The reference paper is "CS²-Collector: A New Approach for Data Collection in Wireless Sensor Networks Based on Two-Dimensional Compressive Sensing." Wang et al. combine temporal and spatial compression to reconstruct a sensor network's physical profile. The paper describes its architecture and experiments and uses a public environmental dataset, making it a useful starting point for RCDD. Related studies help explain where this project fits within existing work. [1, 2]

The central question is whether a gateway can reduce raw telemetry while keeping enough evidence to answer predefined rules without receiver feedback. It must also have enough capacity to send that evidence. Reconstruction error alone cannot answer this question: even a small average error can move a reading across a threshold and change the result of a rule.

1.2 Related work and the project contribution

Earlier studies have already connected load shedding with query semantics. Proof-Infused Streams authenticates answers to stream queries, while runtime verification over out-of-order streams deals with incomplete observations. Technical prior art also describes one-way consistency manifests. Together, these studies establish query-aware reduction, authenticated summaries and UNKNOWN outcomes as existing ideas. [3-6]

RCDD combines these ideas when the gateway accepts an observation. Before allowing irreversible reduction, it records the original rule facts and reserves the capacity needed to send them. Simpler exact summaries that retain the same facts provide a useful basis for comparison. The project implements this combination, although the literature reviewed here does not establish that RCDD is the first system to use it. [P1, P2, P4]

1.3 Scope of this report

Sections 2-4 review the reference paper and identify the research gap. Sections 5-7 introduce the source data, explain how it maps into RCDD and describe the implemented workflow. Sections 8-10 present the new pilot, and Section 11 discusses the findings. Throughout the report, the paper's published results, earlier project tests and new pilot results are kept distinct.

2 Architecture and Method of the Reference Study

CS²-Collector arranges sensor readings in a matrix with dimensions for time slots and sensors. Each sensor compresses its own time series, and a spatial projection determines which sensors report. The base station then applies sparse reconstruction to estimate the full signal from the reduced measurements. [1, Sections 4.1-4.3]

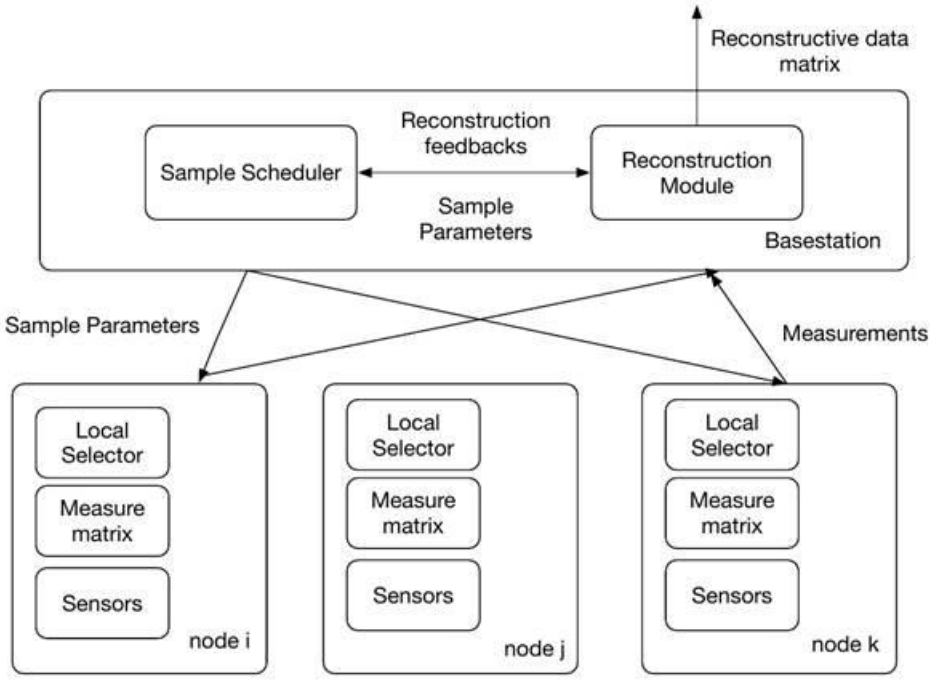


Figure 1 Original CS²-Collector architecture, reproduced from Wang et al. (2016), Figure 1, p. 5, Sensors 16(8), 1318, DOI 10.3390/s16081318, CC BY 4.0. The image is cropped from the publisher PDF.

2.1 Published pipeline

The base-station scheduler sets the sampling parameters, projection matrices and transmission slots. Sensors use these settings to send compressed vectors. To recover the signal matrix, the reconstruction module combines a two-dimensional sparse representation with a Kronecker formulation and ℓ1 optimisation. Figure 1 shows the parameters sent to the sensors and the reconstruction feedback used inside the base station. [1, pp. 4-6]

2.2 Comparison with RCDD

The published architecture has a control downlink to the sensors. In RCDD's research path, contracts are configured locally on both sides, and Fog sends no application feedback to the sender. These operating conditions differ from the reference study; they do not imply that compressive sensing needs continuous feedback. RCDD also produces facts for registered rules instead of reconstructing a physical profile.

This comparison raises two practical questions. Does the reduced information still support a correct monitoring-rule answer? Can the gateway send enough evidence to support that answer when Fog cannot request repairs? The project examines both requirements in addition to the reconstruction question addressed by the reference study.

3 Published Experimental Data and Results

Table 1 outlines Wang et al.'s experiment. The authors use a cleaned portion of the Intel Lab dataset, but do not fully identify the mote IDs, dates, discarded records or random seeds used in their evaluation. [1, Sections 5.1 and 5.3]

Table 1 Published CS² Collector experiment settings

Published setting	Value used in CS²-Collector
Input variables	Temperature, humidity, light and sensor voltage
Matrix for each variable	50 sensors × 50 time slots
Cleaning	Remove erroneous readings; linearly interpolate missing values
Transform and solver	Discrete cosine transform; l1magic reconstruction

Compression study	Removed-element ratio 0.1- 0.9; 30 random-projection trials; mean MSE
Comparators	Temporal 1DCS, spatial 1DCS and joint sparsity model (JSM)
Execution environment	MATLAB 2015b; 2.9 GHz dual-core i5; 16 GB RAM; Mac OS X El Capitan

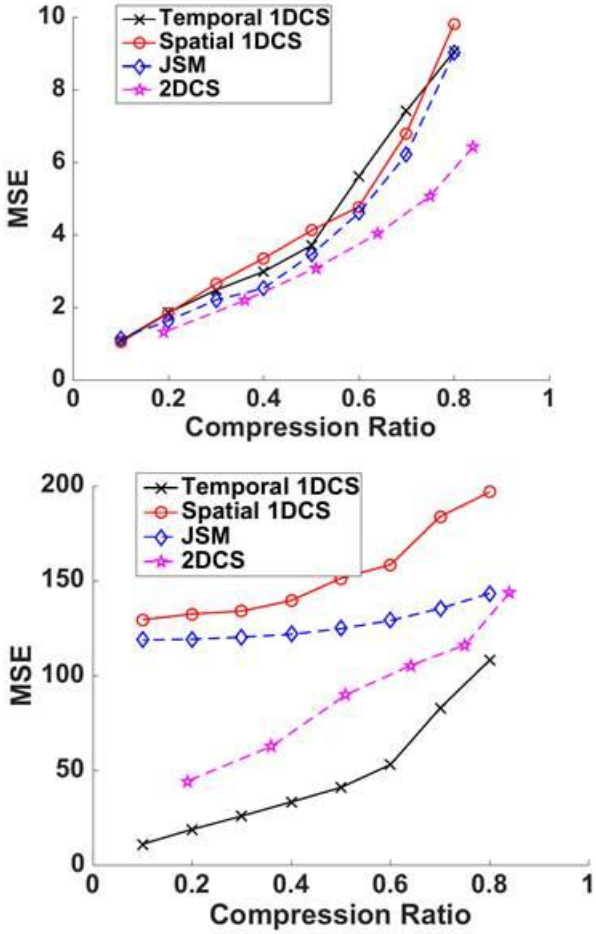


Figure 2 Published results for temperature (left) and light (right), reproduced from Wang et al. (2016), Figures 5 and 11, pp. 10 and 13, DOI 10.3390/s16081318, CC BY 4.0. Both images are cropped from the publisher PDF and keep the original axis scales.

The paper finds lower reconstruction MSE with 2DCS for temperature, humidity and voltage. For light, temporal 1DCS performs best. The authors attribute this difference to weaker spatial sparsity from local lighting and occlusion. Figure 2

keeps the published curves because the paper gives no table of exact MSE values for these results. [1, Figures 5, 7, 9 and 11]

In the paper, compression ratio means the fraction of matrix elements removed. A ratio of 0.9 leaves 10% of the elements; it does not measure the total network bytes saved. The evaluation combines simulations with recorded sensor data. The authors leave experiments on a physical testbed for future work. [1, Equation 14 and Section 6]

4 Research Gap and RCDD Contribution

CS²-Collector measures how accurately a compressed signal can be reconstructed. It does not test whether Fog can answer an exact registered rule after reduction or define what Fog should conclude when evidence is missing. These questions add an application requirement; they do not demonstrate a fault in the published method.

Table 2 Relationship between the research gap and RCDD mechanisms

Question left open by the reference study	Implemented RCDD response	What has been tested
G1 Correct rule decisions after reduction	Count matching, nonmatching and unresolved observations before reducing raw data.	Production rule functions tested on unchanged Intel rows; no comparison with the CS² algorithm.
G2 Accounting for accepted and omitted readings	Durable ingress identity, window scope and signed Rule Coverage Capsules.	Project ledger and verification tests; the archive does not include every physical sample.
G3 Evidence when the receiver cannot request repairs	Set contracts locally, use the forward UDP path and return UNKNOWN when evidence is insufficient.	Earlier local integration tests; this pilot tests rule facts only.

G4 Capacity needed to send the evidence	Reserve storage and future transmission bytes as observations are accepted.	Earlier synthetic tests at matched budgets; no Intel byte-cost measurement.

4.1 How RCDD addresses the gap

Before the gateway reduces accepted raw data, RCDD stores the facts required by the registered rules and reserves resources to send them. It then packages the facts in signed capsules for Fog. Table 2 breaks this process into four needs: preserve rule meaning, identify accepted observations, work without receiver repair and account for the resources used by the evidence. [P1-P3]

4.2 What the project can claim

The implementation can be checked for correct rule decisions and for its behavior when evidence is insufficient. Whether dynamic reservations produce more useful decisions per byte than compact exact summaries is a separate question. The synthetic results show no advantage for the reservation approach in that comparison. [P4]

The proposed novelty lies in combining these mechanisms around the gateway's admission decision. Signatures, sufficient statistics and three-valued logic are existing tools. Preparing facts for registered rules also has a tradeoff: those facts cannot reconstruct the full signal and may not answer a new query once the required raw inputs have been discarded.

5 Original Dataset and Actual Input Records

The pilot draws on the public Intel Lab archive cited by the reference study. Collected in the 2004 Intel Berkeley deployment, its records contain date, time, epoch, mote ID, temperature, relative humidity, light and voltage. The four measured variables use degrees Celsius, percent, lux and volts, respectively. [1, 2]

5.1 Source file and data quality

Table 3 Measured profile of the downloaded public archive

Property of the downloaded file	Measured value
Compressed size	34,422,518 bytes
All source lines	2,313,682
Lines with eight fields	2,219,803
Lines with another field count	93,879 lines; no automatic repairs
Archive integrity	Full gzip file read successfully; compressed-file SHA-256 retained

5.2 Original sensor observations

Table 4 Unmodified Intel Lab source observations

Source line	Mote	Time	Epoch	Temp °C	RH %	Light lux	Volt V
3055	1	00:01:57.13085	5648	18.4498	43.1191	43.24	2.67532
46422	2	00:00:22.429139	5645	18.8712	43.4524	128.8	2.63964
93509	3	00:01:50.626164	5648	18.7536	42.7185	48.76	2.65143
140558	4	00:01:00.302927	5646	19.1162	42.5513	97.52	2.61639
182777	6	00:00:59.282931	5646	18.6752	42.5847	114.08	2.61639

Table 4 gives each selected mote's first complete record on 1 March 2004. Values retain the source file's decimal precision, and source line numbering starts at one. These are original public records. The paper's authors do not separately disclose the cleaned 50 × 50 partition used in their experiment. [2, E1]

5.3 Why this data fits RCDD

The records fit RCDD's environmental telemetry format. Native fields already support temperature and humidity, and named measurements can hold light and voltage. Only temperature and humidity enter the pilot's rules. All five motes are treated as environmental devices, with their readings kept as environmental observations rather than converted into synthetic CPU, network, security or industrial metrics.

Peter Bodik, Wei Hong, Carlos Guestrin, Sam Madden, Mark Paskin and Romain Thibaux collected the dataset using hardware supplied by Intel Berkeley. The dataset page allows reuse with acknowledgment of their work. [2]

6 Mapping the Data into the Project

Table 5 Mapping original data fields into RCDD

Original field or property	How the pilot handles the field
Mote ID	Stable device ID such as intel-mote-01; all five devices are environmental.
Temperature and humidity	Keep finite values unchanged in the native fields. Humidity must meet the native 0-100 check.
Light and voltage	Can use measurements.light and measurements.voltage. Neither is a rule input in this pilot.
Source date and time	Keep the original time without assigning a timezone. A separate replay clock assigns rule windows.
Source epoch	Keep it as the mote sequence, separate from Unix time and the RCDD ledger epoch.

Missing or malformed records	Leave gaps and values unchanged. Selected rows must pass the native payload validator.

6.1 Selecting the records

The pilot uses 1 March 2004 and selects the first five mote IDs in ascending order with complete eight-field records: 1, 2, 3, 4 and 6. Mote 5 has no complete records in that form. All 8,709 selected rows pass the native telemetry validator, with no rows removed or filled in at that stage. The subset has no exact duplicate rows or repeated epochs within a mote. Both the selection and the thresholds were fixed before evaluating rule outcomes. [E1]

Table 6 Selected records and observed sampling gaps

Mote	Original rows	Median observed gap s	Largest observed gap s
1	1,921	33.522	295.733
2	1,724	35.161	351.171
3	1,976	33.226	240.163
4	1,749	35.052	324.521
6	1,339	47.638	333.183

6.2 Differences in data preparation

Wang et al. discard erroneous readings and use linear interpolation to fill gaps in their 50 × 50 matrices. This pilot retains the observed rows because estimated readings would change the rule counts. A complete replay window therefore accounts only for accepted archive rows and cannot prove that every physical sample was recorded. The source timestamps have no specified timezone. [1, 2, E1]

The frozen RCDD contract supports up to 16 devices and 32 rules, enough for this five-mote pilot. Testing 50 motes natively would require separate bounded contracts or a reviewed extension. Although both studies use the same public archive, the pilot does not reproduce the paper’s undisclosed cleaned partition. [P1]

7 RCDD Architecture and Integrated Workflow

Workflow pipeline - integrated native system

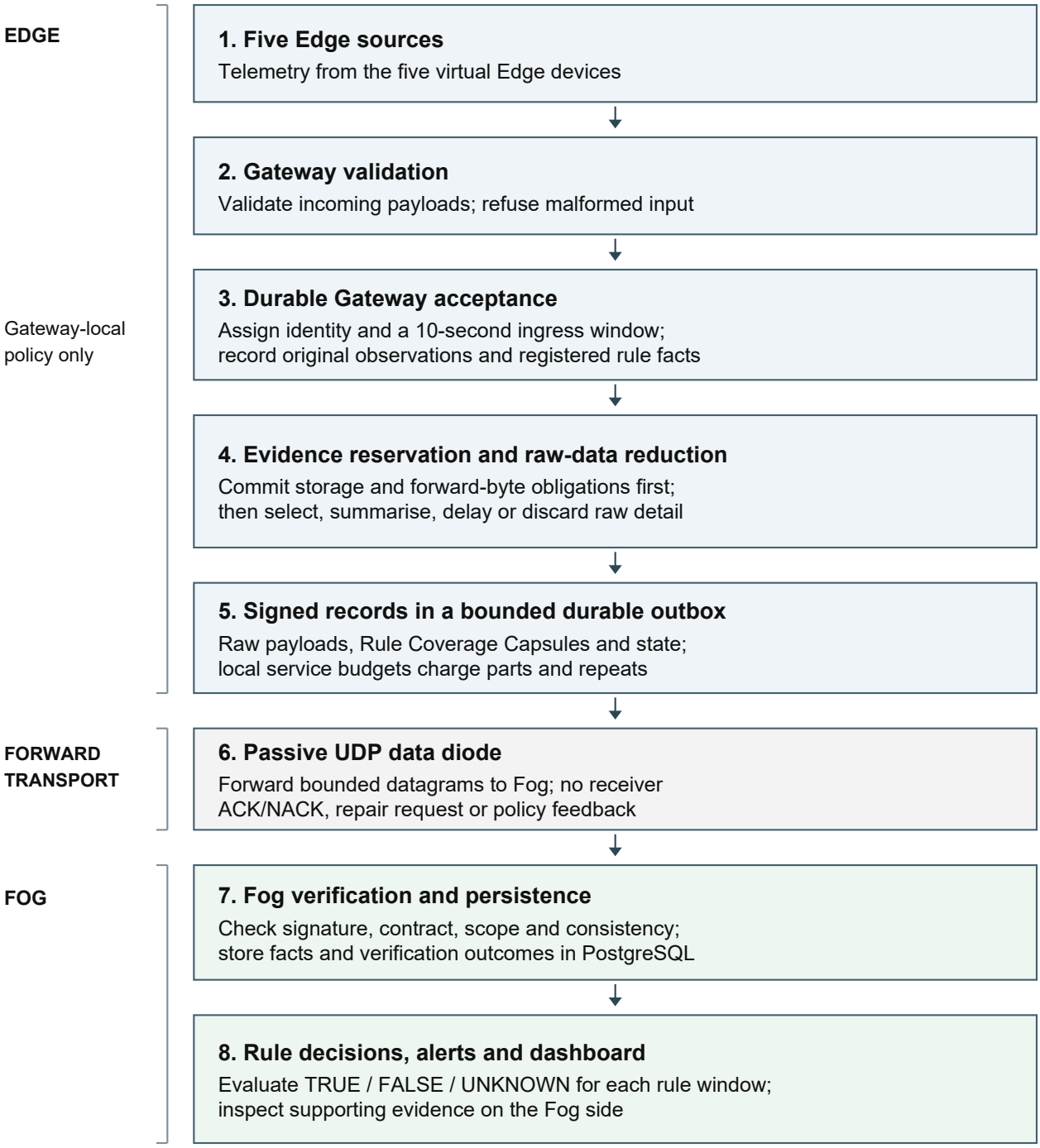


Figure 3 Native RCDD workflow pipeline. Original rule facts and evidence capacity are maintained before raw-data reduction. The Intel pilot calls the production fact and rule functions only; it does not run this full transport pipeline.

7 RCDD Architecture and Integrated Workflow (continued)

7.1 Before raw data is reduced

The Edge gateway validates incoming telemetry before accepting it. Once a reading is durably accepted, the gateway gives it an original identity and a ten-second ingress window and updates the registered rule facts. It checks storage and reserves future capacity to send the evidence. Only then can raw detail be selected, summarised, delayed or discarded. If these obligations cannot be met, the gateway rejects admission or records an explicit limitation. [P1, P2]

7.2 Sending the rule evidence

Each signed Rule Coverage Capsule (RCC) identifies its gateway, contract, ledger epoch, device and window. It carries the ingress range, condition counts, reduction record, limitations, input commitment and predecessor digest. A bounded durable outbox sends the signed parts through the passive UDP diode, reusing the same signed bytes on repeated attempts. Fog sends no application acknowledgments, retransmission requests or policy updates back to the research sender. [P2, P3]

7.3 Fog decisions and rule semantics

Fog checks signatures, format, contract, identity and consistency, then stores facts and verdicts in PostgreSQL. For each condition, m counts matches, n counts nonmatches and u counts unresolved observations. Within a valid closed scope, the match count is between m and m + u. COUNT_GE(K) is TRUE if $m \geq K$. FALSE requires sufficient observations and $m + u < K$; otherwise the answer is UNKNOWN. EXISTS uses $K = 1$. AND and OR follow strong Kleene logic, allowing a definite result when one branch supplies enough evidence to settle the answer. [P1, P3]

7.4 Phase integration and trust assumptions

The twelve phases build on the project's earlier platform. N0-N1 define the contract and rule semantics, and N2-N4 add durable facts, signing and evidence-aware reduction. N5-N8 implement the forward path, Fog storage, rules and inspection. N9-N11 test failures and compare the results. This remains a software boundary, without hardware-enforced one-way transfer. A signature confirms what the trusted gateway states; it cannot prove the physical truth of measurements that the receiver has not seen. [P1-P4]

8 Methodology of the Real Data Pilot

The pilot evaluates rules offline against the original dataset rows. It calls the production condition_facts and evaluate_rule functions, validates native telemetry and compares the answers with the original-trace oracle. Gateway admission, reservations, signing, sockets, the diode and PostgreSQL are outside this test. [E1]

8.1 Rules and replay timing

Each mote has four rules: EXISTS and COUNT_GE(3) for temperature ≥ 23 °C, plus EXISTS and COUNT_GE(3) for humidity ≥ 40%. Two further rules apply AND and OR to the temperature EXISTS answers from motes 1 and 2, bringing the total to 22 rules. The thresholds are fixed examples for this experiment. They are neither event labels supplied by Wang et al. nor validated safety limits.

Table 7 Time mappings and rule window populations

Mapping	Source interval and observations	Rule windows and opportunities per arm
Primary 30× modeled replay	Full fixed day; 8,709 rows. Each 300 source seconds maps to 10 modeled ingress seconds.	288 windows × 22 rules = 6,336
Natural cadence sensitivity	First two source hours; 684 rows. Source elapsed time maps 1:1 to ingress elapsed time.	720 windows × 22 rules = 15,840

The replay clock starts at a synthetic, aligned UTC origin. It does not assign a timezone to the source records or recreate their actual gateway arrival times. Both mappings use the contract's ten-second windows. The natural-cadence test uses part of the primary dataset, so it checks timing sensitivity rather than providing an independent repeat.

8.2 Comparison arms

Table 8 Pilot comparison arms

Arm	What changes in this arm
Exact facts	Build exact condition facts from the original rows and evaluate the rules.
Compact exact control	Use the same facts and evaluator, which must give matching answers. Encoding and byte cost are not measured.
Fixed evidence omission	Remove all facts from every fifth window. This fixed schedule does not represent measured packet loss.
Unsafe mean substitution	Replace each finite reading with its window mean, keeping the original count. This deliberately unsafe control does not implement CS²-Collector.

8.3 Metrics and test suite

Correct definite yield measures the share of all rule-window opportunities that receive a correct TRUE or FALSE answer. Its denominator includes reference UNKNOWN cases. Wrong definite answers and UNKNOWN outcomes are counted separately. A verifier written independently of the production evaluator recomputes 20,160 condition facts and checks all 88,704 result rows, finding no disagreements. The results are deterministic and involve related rules and windows, so they are not independent trials for confidence intervals or significance tests. [E1]

9 Results Using the Intel Lab Dataset

9.1 Primary modeled replay

All 288 primary windows contain observations across the selected motes. Complete exact facts let every rule be decided within its tested scope, giving each arm 6,336 opportunities. Table 9's 100% yield depends on complete facts and the 30× time mapping. It is not a measure of native real-time performance. [E1]

Table 9 Results under the 30× modeled replay

Arm	TRUE	FALSE	UNKNOWN	Wrong definite	Correct yield
Exact facts	2,545	3,791	0	0	100.00%
Compact exact control	2,545	3,791	0	0	100.00%
Fixed evidence omission	2,034	3,026	1,276	0	79.86%
Unsafe mean substitution	2,492	3,844	0	63	99.01%

The exact-fact and compact-control arms return the same answers because they share the facts and evaluator. This does not test dynamic admission or byte cost. Removing all facts from every fifth window withholds evidence from 58 of the 288 windows, producing 1,276 UNKNOWN answers and no wrong definite answers. The omitted windows account for 20.14% of the total. That proportion comes from the fixed experiment schedule, rather than a measured 20% rate of network loss.

The unsafe mean control returns 63 wrong answers: FALSE in 58 cases where the reference is TRUE, and TRUE in five cases where it is FALSE. These errors follow from the deliberately unsafe substitution in this pilot. They say nothing about CS²-Collector's reconstruction performance. Section 10 works through two examples using the original source rows.

9.2 Results at the natural sampling cadence

At the original cadence, only 302 of the 720 windows in the two-hour replay contain observations from the selected motes. Gaps in individual devices' readings leave many rule windows without enough evidence. Each arm has 15,840 opportunities. [E1]

Table 10 Results under the natural cadence sensitivity

Arm	TRUE	FALSE	UNKNOWN	Wrong definite	Correct yield
Exact facts	684	2,356	12,800	0	19.19%
Compact exact control	684	2,356	12,800	0	19.19%
Fixed evidence omission	552	1,902	13,386	0	15.49%
Unsafe mean substitution	684	2,356	12,800	0	19.19%

At this cadence, exact facts give a correct definite yield of 19.19%. The unsafe mean control happens to match the reference in these sparse windows, but that agreement does not make the method sound in general. Withholding evidence lowers the yield further. The contrast with the primary replay shows why those results cannot be applied directly to a native deployment with ten-second monitoring windows.

10 Rule Decisions on Original Records

The examples below use unchanged Intel archive records from five-minute source bins in the primary replay. The original readings give the reference answer in each case. Replacing them with the window mean then shows how this deliberately unsafe representation can change the rule's result. [2, E1]

10.1 A humidity match hidden by the mean

For mote 1 on 1 March 2004, consider the 07:10-07:15 source window. The rule asks if any humidity reading reaches 40%. Table 11 gives the original records.

Table 11 Original humidity observations for mote 1

Source line	Mote epoch	Humidity %
3530	6506	40.0268
3531	6508	39.6539
3532	6510	39.5181
3533	6511	39.4502
3534	6512	39.4162
3535	6513	39.4502
3536	6514	39.3483

The first of these seven readings reaches the threshold, making EXISTS TRUE. Their mean is 39.551957%, rounded to six decimals. Replacing every reading with that mean makes the unsafe answer FALSE. Retaining the exact matching count, m = 1, keeps the correct positive result without requiring Fog to store all seven raw readings.

10.2 A count inflated by the mean

For mote 6 on 1 March 2004, the source window is 10:20-10:25. The rule requires at least three temperature readings at or above 23 °C. Table 12 lists the four original readings in that window.

Table 12 Original temperature observations for mote 6

Source line	Mote epoch	Temperature °C
183262	6885	22.9676
183263	6889	22.997
183264	6892	23.0362
183265	6893	23.0166

Two readings meet the threshold, so COUNT_GE(3) is FALSE. The mean is 23.00435 °C. Replacing all four readings with it creates four apparent matches and changes the

unsafe answer to TRUE. Exact facts keep the original two matches and two nonmatches, preserving the correct count.

These examples explain why the project keeps facts tied to the rule. An average may capture a signal's general level while losing the information needed for a correct answer. This finding applies to the recorded observations only; it does not cover samples missing from the archive.

11 Discussion Limitations and Conclusion

11.1 What the findings show

The pilot examines the public data used by CS²-Collector from the perspective of rule decisions. Exact facts retain the tested counts and answers, and the evaluator returns UNKNOWN when scheduled evidence is withheld. The examples show why keeping an average alone may be insufficient. This supports preserving rule evidence, without evaluating the paper's compression algorithm. [1, E1]

11.2 Evidence from the integrated system

The retained N11 evidence includes a 694-test regression and seven native scenarios covering reduction, positive rules, cross-device rules, raw-data loss, evidence loss, tampering and recovery. These tests exercise durability, signing and the configured forward path using the earlier synthetic workload. Their findings are separate from the offline Intel pilot and do not establish that the full system has been deployed with real data. [P3, P4]

The earlier N10 study compares arms at matched byte budgets over 840 model trials. RCDD ties matched-format summaries in all fourteen case means. Compact exact signed summaries achieve higher mean decision yield in twelve cases and tie in two. These settings therefore show no yield advantage from the reservation mechanism. The Intel pilot does not measure bytes and offers no new evidence to change that finding. [P4]

11.3 Limits and next experiment

The pilot covers one day, five motes and illustrative environmental thresholds. The archive lacks independent real-world event labels and does not contain every physical observation. Faster replay fills more windows, while the natural cadence leaves many rules undecided. The experiment does not measure full-signal reconstruction, new

unregistered rules, native packet cost, energy or long-run operation. It also relies on trust in the gateway, which could sign an incorrect statement.

The next experiment should run the Intel records through native admission, durable reservations, signing and Fog. RCDD and compact exact summaries should then be compared under the same measured total byte budget. This needs consistent UTC and monotonic clocks, a tick at every ten-second boundary and explicit retention of missing windows. The current 256-capsule signing-history limit needs review before an endurance run. Comparing CS² would require reproducing it on a disclosed common partition. [P1, P4]

11.4 Conclusion

RCDD adds a practical question to the reference study's focus on reconstruction: what can a receiving application safely decide after raw data is reduced? The implementation combines rule facts recorded before reduction with an advance commitment to send their evidence. The Intel records show the value of those facts and the strong effect of window timing. The findings support the tested behavior, but do not establish better performance than the strongest simpler control or show that the approach is globally original.

References

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https://patents.google.com/patent/US11641327B2/en

Project and experiment references

P1-P4 refer to retained project sources, and E1 refers to the new offline Intel experiment. Appendix A lists their source paths and provenance. The external primary sources were checked on 19 October 2026.

Figure license

Figures 1 and 2 use cropped portions of the reference paper under CC BY 4.0. They retain the original labels and result scales, with attribution beside each figure. License: Creative Commons Attribution 4.0

Appendix A Reproducibility and Evidence

This appendix lists the evidence for the new pilot and the existing implementation. Paths are relative to the two roots below. The pilot reads the production source without changing the deployed project.

A1 New data and experiment evidence

Report workspace C:/Users/Rithvik Kumar/Desktop/fog_edge_prj

Evidence folder report_work/literature_evidence/

[E1] The pilot uses run_intel_semantic_pilot.py and verify_intel_semantic_pilot.py. The intel_semantic_pilot_v1/ folder stores the contract, manifest, source-quality counts, original rows, condition facts, verdict CSV, summary, counterexamples and independent checks.

Dataset SHA-256

d99288c8f406ca6604d359ceaa0d8adfffa79e7095061a1e27dc4399f48c7225

Contract SHA-256

4c711913a8b5c2350926f28eff5e6967e31d2c345b4686eeefc168ddb3214988

The manifest records Python 3.12.6, Pydantic 2.13.5 and hashes of the production modules and pilot script. To rerun the pilot, use the project virtual environment with -B and supply --project, --dataset and --output, selecting a new output child folder. The verifier takes no arguments and checks the fixed intel_semantic_pilot_v1 folder.

A2 Implementation sources

Project root C:/Users/Rithvik Kumar/Desktop/fogedge/software-data-diode

[P1] Contract and rule semantics

docs/novelty-roadmap/reports/N0-contract-v1.md

shared/coverage/contracts.py; evaluation.py; oracle.py; models.py

[P2] Durable admission and evidence signing

shared/coverage/ledger.py; admission.py

docs/novelty-roadmap/reports/N2.md; N3.md

[P3] Forward transport and Fog processing

docs/novelty-roadmap/reports/N5.md; N6.md; N7.md; N8.md

[P4] Comparative experiments and native tests

docs/novelty-roadmap/reports/N9.md; N10.md; N11.md

experiments/results/coverage/n10-closeout-analysis-20261015/statistics.csv

docs/novelty-roadmap/evidence/N11/closeout-regression.xml

A3 Provenance of tables and figures

Section 3's settings and Figure 2's plots come from Wang et al. [1]. Sections 5, 6 and 10 draw on the public archive [2] and retained source-line records [E1]. Section 9 reports the new pilot's results, separate from those of Wang et al. Figure 3 depicts the integrated project path. In the Word source, text and data tables remain editable, while the source figures are included as images.